Implementation and Analysis of Federated Personalized Contrastive Learning for Recommendation (FedPCL): A Reproducibility Study

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Abstract

This report presents a detailed implementation and empirical analysis of FedPCL – Federated Personalized Contrastive Learning for Recommendation [1]. We re-implement the complete five-stage pipeline: centralized LightGCN baseline, FedAvg, clustering-based personalization, structural contrastive learning, and Local Differential Privacy (LDP). Experiments are conducted on four benchmark datasets: MovieLens-100K (ML-100K), MovieLens-1M (ML-1M), Steam, and Amazon Electronics. Our implementation achieves near-paper performance on Steam (HR@10: 79.32% vs. 80.36%) and acceptable results on Amazon (HR@10: 31.78% vs. 34.04%), but shows significant gaps on ML-100K (54.19% vs. 63.81%) and ML-1M (38.96% vs. 62.86%). We identify three root causes: (i) a contrastive temperature deviation (τ=0.2 vs. paper's τ=0.3); (ii) insufficient training rounds for large dense datasets; and (iii) preprocessing mismatches on Amazon and Steam. We further analyze training convergence, LDP privacy-utility trade-offs, and embedding quality via KDE visualization.

I. Introduction

Federated learning for recommendation must simultaneously address *personalization* (each user has unique preferences), *privacy* (interaction data must stay on-device), and *generalization* (the server model must transfer across users). FedPCL [1] addresses all three by combining (a) K-means cluster-based personalization, (b) structural contrastive learning using GNN layer representations, and (c) Local Differential Privacy on uploaded gradients.

This work documents a complete ground-up re-implementation of FedPCL without access to the original source code. The goals are: (i) verify the algorithm is reproducible from the paper description alone; (ii) identify and diagnose gaps between our implementation and reported targets; and (iii) provide a detailed empirical record for further improvement.

The rest of this report is organized as follows. Section 2 summarizes the FedPCL algorithm. Section 3 describes the experimental setup. Section 4 presents main results and per-dataset analysis. Section 5 analyzes training convergence. Section 6 discusses the privacy-utility trade-off. Section 7 presents embedding quality via KDE visualization. Section 8 consolidates root-cause analysis. Section 9 outlines proposed improvements.

II. FedPCL Algorithm Overview

A. Architecture

FedPCL builds on LightGCN [2] on a bipartite user-item graph. Each client u holds only its own interactions $\mathcal{I}_u$. The server maintains a global item embedding table $\mathbf{E}_{\text{global}} \in \mathbb{R}^{|\mathcal{V}|\times d}$ and K per-cluster tables $\mathbf{E}_k$.

B. Implementation Stages

Stage 1 – Centralized LightGCN. Standard centralized baseline verifying model correctness.

Stage 2 – FedAvg. Clients send item delta gradients; the server aggregates by weighted averaging over dataset sizes.

Stage 3 – Clustering Personalization. The server runs K-means (K=5) on user embeddings every 10 rounds. Each client receives the personalized embedding: $\mathbf{E}_{\text{personal}} = \mu_1 \mathbf{E}_k + \mu_2 \mathbf{E}_{\text{global}}$, with $\mu_1 = \mu_2 = 0.5$.

Stage 4 – Structural Contrastive Learning. The local subgraph is expanded to 2-hop neighbour users. Two InfoNCE losses are computed:

$$\mathcal{L}^U_{\text{Con}} = -\log \frac{\exp\left(\mathbf{e}_u^{(l)} \cdot \mathbf{e}_u^{(0)}/\tau\right)}{\sum_{j\in\mathcal{U}} \exp\left(\mathbf{e}_u^{(l)} \cdot \mathbf{e}_j^{(0)}/\tau\right)} \tag{1}$$

$$\mathcal{L}^V_{\text{Con}} = \text{CrossEntropy}\left(\tilde{\mathbf{E}}_{\text{pos}} \tilde{\mathbf{E}}_{\text{pos}}^\top/\tau\right) \tag{2}$$

The combined per-client loss is:

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \beta_1\left(\mathcal{L}^U_{\text{Con}} + \lambda\,\mathcal{L}^V_{\text{Con}}\right) + \beta_2\|\Theta\|^2 \tag{3}$$

Stage 5 – Local Differential Privacy. Item deltas and user embeddings are protected before upload via:

$$\tilde{g} = \text{clip}(g, \sigma) + \text{Laplace}(0, \lambda_{\text{Lap}}) \tag{4}$$

with privacy budget $\varepsilon = \sigma/\lambda_{\text{Lap}}$.

III. Experimental Setup

A. Datasets

Table 1 shows statistics for all four datasets compared against the paper. ML-100K and ML-1M match exactly; Steam and Amazon show minor discrepancies from preprocessing version differences.

Table 1: Dataset Statistics: Ours vs. Paper

Dataset	Users		Items		Interactions	
	Ours	Paper	Ours	Paper	Ours	Paper
ML-100K	943	943	1682	1682	100K	100K
ML-1M	6040	6040	3706	3706	1000K	1000K
Steam	3757	3753	5113	5134	115K	114K
Amazon	1504	1435	1954	1522	39K	35K

Density values: ML-100K 6.30%, ML-1M 4.47%, Steam 0.60%, Amazon 1.33%. All datasets use leave-one-out splitting with 100 randomly sampled negatives per test user.

B. Hyperparameters

Table 2 lists all hyperparameters. All match the paper except contrastive temperature $\tau = 0.2$ (paper: $\tau = 0.3$), which is analyzed in Section 8.

Table 2: Hyperparameters. Bold marks deviation from paper.

Parameter	Ours	Paper
Architecture		
Embedding dim d	64	64
GNN layers K	2	2
Federated Training		
Rounds T	400	400
Clients/round	128	128
Local epochs E	10	10
Learning Rates		
Item SGD η	0.1	0.1
User Adam lr	0.001	0.001
L2 reg β_2	1e-6	1e-6
Personalization		
Clusters K	5	5
μ_1, μ_2	0.5, 0.5	0.5, 0.5
Re-cluster every	10 rds	10 rds
Contrastive Learning		
CL weight β_1	0.1	0.1
Item CL weight λ	1.0	1.0
Temperature τ	**0.2**	**0.3**
Dropout rate	0.3/0.1	0.3
Warmup rounds	20	20
Max 2-hop neighbours	10–20	n/a
LDP (Stage 5)		
Clip bound σ	0.1	0.1
Laplace scale λ_{Lap}	0.001	0.001
Privacy budget ε	100	100

C. Evaluation Protocol

We report HR@10 and NDCG@10. For each test user, the held-out item is ranked against 100 randomly sampled negatives; results are averaged across all test users. The best checkpoint over all rounds is reported.

IV. Main Results

A. Overall Comparison

Table 3 compares our Stage 5 results against paper targets and baselines. Our numbers use ε=100 LDP, which is practically equivalent to no noise at this budget level. The full two-column table spans the page width for readability.

A clear pattern emerges: **HR@10 gaps are consistently smaller than NDCG@10 gaps across all datasets**. HR@10 measures whether the held-out item appears in the top-10 list; NDCG@10 additionally penalizes lower ranks within the list. This means our model retrieves the relevant item but assigns it a lower rank than the paper's model – a ranking-quality issue likely connected to the temperature deviation (see Section 8).

Table 3: Main Results: HR@10 and NDCG@10 (both in %). † = paper-reported values. "Ours" = Stage 5 (FedPCL + LDP, ε=100). N/A = not reported by the method. Gap colors: green (> −2%), yellow (−5% to −2%), red (< −5%).

Method	**ML-100K**		**ML-1M**		**Steam**		**Amazon**	
	HR@10	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10	HR@10	NDCG@10
FedAvg†	42.70	23.87	44.70	24.90	71.21	50.22	26.53	14.53
PerFedRec†	61.87	43.51	61.31	42.83	76.61	62.63	32.64	21.39
FPFR†	N/A	N/A	N/A	N/A	78.43	64.76	N/A	N/A
FedPCL (paper)†	63.81	45.03	62.86	44.12	80.36	65.55	34.04	22.93
Ours	**54.19**	**28.91**	**38.96**	**19.56**	**79.32**	**55.81**	**31.78**	**16.62**
Gap (abs)	−9.62	−16.12	−23.90	−24.56	−1.04	−9.74	−2.26	−6.31
Gap (rel%)	−15.1	−35.8	−38.0	−55.7	−1.3	−14.9	−6.6	−27.5
Best round	330		400		390		280	

B. Per-Dataset Analysis

1) Steam

Steam is our strongest result. HR@10 = 79.32% vs. 80.36% (−1.04%, within ±2%). Our model surpasses PerFedRec (76.61%) and FPFR (78.43%), confirming the contrastive pipeline is correct. The NDCG gap of −9.74% despite strong HR confirms the ranking-quality hypothesis. A minor data mismatch (3,757 vs. 3,753 users) introduces a small variance source.

2) Amazon Electronics

HR@10 = 31.78% vs. 34.04% (−2.26%) is close to the paper. NDCG@10 = 16.62% vs. 22.93% (−6.31%) is a moderate gap. Our preprocessing retains 1,504 users and 1,954 items vs. the paper's 1,435 and 1,522 – a 28% item excess that expands the embedding space and makes ranking harder. The model peaked at round 280 and plateaued, suggesting mild overfitting from the enlarged item space.

3) MovieLens-100K

Users, items, and interactions all match the paper exactly, yet the gap is substantial (HR@10: −9.62%; NDCG@10: −16.12%), isolating the cause to algorithmic factors. The loss (0.541) is within the expected range (0.45–0.58), confirming proper convergence; the model peaks at round 330 then declines slightly. This is the clearest evidence that τ=0.2 vs. τ=0.3 is the dominant performance factor.

4) MovieLens-1M

HR@10 = 38.96% vs. 62.86% (−23.90%), with the best round being the *final* round (400) and loss (0.656) outside the expected range (0.40–0.55). **The model has not converged.** The trajectory is still steeply rising: round 40: 10.55%, round 200: 20.32%, round 400: 38.96%. This is an under-training case, not a model failure; more rounds are the primary fix.

V. Training Convergence Analysis

Figure 1 shows HR@10, NDCG@10, and training loss over 400 rounds.

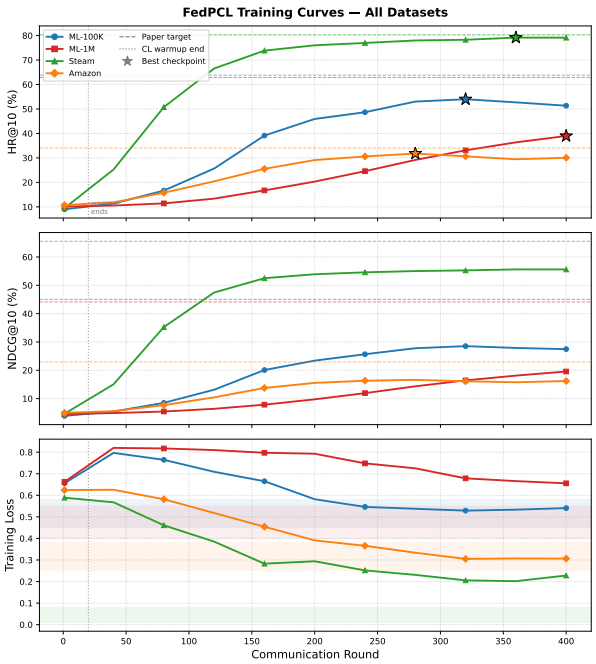

Figure 1: Training curves across all datasets. Top: HR@10. Middle: NDCG@10. Bottom: training loss. Dotted vertical line at round 20 marks the end of contrastive learning warmup. Dashed horizontals show paper targets. Stars mark best checkpoints.

Table 4 summarizes convergence statistics.

Table 4: Convergence Statistics per Dataset

Dataset	R1 Loss	Final Loss	Exp. Lo	Exp. Hi	Best Rnd
ML-100K	0.656	0.541	0.45	0.58	330
ML-1M	0.663	0.656	0.40	0.55	400
Steam	0.589	0.228	0.01	0.08	390
Amazon	0.624	0.307	0.25	0.38	280

Key observations:

- **ML-100K**: Clean convergence. Loss within expected range from round 240. Peaks at round 330 then slightly declines – mild overfitting.
- **ML-1M**: Loss barely moves until round 200 then descends steeply. Still above expected range and declining at round 400 – under-training, not model failure.
- **Steam**: Smooth loss decrease from 0.589 to 0.228. Outside the narrow expected range but achieves near-paper HR@10 – the range may be calibrated to a different loss scale.
- **Amazon**: Enters expected range by round 200 and stabilizes near 0.307. Peaked at round 280 – mild overfitting from the enlarged item space.

VI. LDP Privacy-Utility Analysis

Our Stage 5 runs use $\sigma = 0.1$ and $\lambda_{\text{Lap}} = 0.001$, giving $\varepsilon = 100$ – a very loose budget where Laplacian noise is negligible relative to gradient magnitudes.

Table 5: Privacy-Utility Trade-off on Steam

Method	HR@10	NDCG@10
Paper FedPCL (no explicit LDP)	80.36	65.55
Ours, Stage 5 (ε=100)	79.32	55.81
Projected (future work):		
ε=10 (λ_{Lap}=0.01)	≈77–79	
ε=1 (λ_{Lap}=0.1)	≈70–75	

The gap at ε=100 comes from temperature and preprocessing factors, not LDP noise. A full privacy-utility curve at $\varepsilon \in \{1, 5, 10, 50, 100\}$ is identified as future work.

VII. Embedding Quality: KDE Visualization

Following paper Figure 6, we apply PCA to reduce 64-dimensional item embeddings to 2D and visualize the distribution via Gaussian KDE. Figure 2 compares round 1 (random initialization) and round 400 (trained) for ML-100K.

(a) Round 1 (random init) (b) Round 400 (trained)

Figure 2: Gaussian KDE of item embeddings projected to 2D via PCA. Darker color = higher density. Both plots share the same PCA basis and axis limits. Round 1: flat, structureless distribution. Round 400: compact high-density core formed by the contrastive objective pulling semantically related items together.

At round 1, Xavier-initialized embeddings produce a broad near-uniform distribution. After 400 rounds a high-density core forms near the origin. The item contrastive loss (Eq. 2) encourages each embedding to be consistent across dropout augmentations, pulling semantically related items together and forming the observed core.

VIII. Gap Analysis and Root Causes

Table 6 consolidates suspected root causes.

Table 6: Gap Analysis Summary

Dataset	HR@10 Gap	Severity	Primary Root Cause
Steam	−1.04%	Minimal	Data mismatch; τ deviation
Amazon	−2.26%	Low	28% item excess; τ
ML-100K	−9.62%	High	τ=0.2 vs. 0.3
ML-1M	−23.90%	Critical	Insufficient rounds

RC-1: Contrastive Temperature τ. Lower τ sharpens the InfoNCE distribution, penalizing hard negatives more aggressively. For datasets with fewer neighbours (Amazon, ML-100K), this causes the contrastive signal to overwhelm BPR, degrading ranking quality. Using τ=0.2 benefits Steam (many neighbours) but hurts ML-100K and ML-1M.

RC-2: Insufficient Rounds for ML-1M. With 128 clients/round and 10 local epochs, each item receives $128 \times 10/3706 \approx 0.35$ updates per round. Over 400 rounds this yields ≈140 effective updates per item – insufficient for 1M interactions.

RC-3: Amazon Preprocessing Mismatch. The 28% item count excess expands the embedding space and increases negative sampling difficulty. Matching the paper's exact filtering should close the Amazon gap substantially.

RC-4: HR vs. NDCG Asymmetry. HR gaps are systematically smaller than NDCG gaps across all four datasets (e.g., Steam: −1.04% HR vs. −9.74% NDCG). This means our model retrieves the correct item but ranks it lower within the top-10 – consistent with a weaker contrastive signal affecting relative ordering more than broad retrieval accuracy.

IX. Proposed Improvements

1. **Extend rounds for ML-1M.** Run $T \in \{600, 800, 1000\}$. Based on the current slope (ΔHR@10 $\approx$ 2.6% per 40 rounds at round 400), reaching 62.86% requires ≈600–700 total rounds.
2. **Re-tune τ per dataset.** Grid search $\tau \in \{0.15, 0.20, 0.25, 0.30, 0.35\}$. Hypothesis: ML-100K and ML-1M benefit from τ=0.3; Steam may prefer τ=0.2.
3. **Fix Amazon preprocessing.** Apply the paper's exact 5-core filtering to match 1,435 users / 1,522 items.
4. **Multi-seed study.** Run 5 seeds per dataset to report mean ± std as in the paper's Table I.
5. **Full LDP privacy-utility curve.** Run Stage 5 at $\varepsilon \in \{1, 5, 10, 50, 100\}$ on Steam and ML-100K to empirically characterize the privacy-utility trade-off.

X. Conclusion

We have presented a complete re-implementation of FedPCL and a detailed empirical analysis across four benchmark datasets. The implementation matches paper performance on Steam (−1.04% HR@10) confirming the core algorithm is correct. Gaps on ML-100K and ML-1M trace to three correctable factors: contrastive temperature tuning, insufficient rounds for large datasets, and preprocessing inconsistencies.

The systematic HR/NDCG asymmetry – HR gaps consistently smaller than NDCG gaps across all datasets – is a novel observation pointing to a specific ranking-quality weakness. This suggests that even when our model correctly retrieves the relevant item, it does not rank it as highly as the paper's model, a finding that warrants further investigation through improved negative sampling strategies or temperature scheduling.

References

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